A complete characterization of discretizable continuous frames

Literature answer to the discretization problem in arXiv:math/0410571

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Status: literature-already-answered, full characterization. A measurable family can be sampled to a discrete Hilbert frame exactly when it is an essentially bounded continuous frame for some positive σ-finite measure. Hence every bounded continuous frame is sampleable, but arbitrary unbounded continuous frames need not be.

Source problem

A continuous frame $\Psi : X \to H$ with respect to a measure μ satisfies

$$A\|h\|^2 \leq \int_X |\langle h, \Psi(t)\rangle|^2 \, d\mu(t) \leq B\|h\|^2 \qquad (h \in H).$$

The Ali–Antoine–Gazeau discretization problem asks for conditions under which one may choose a countable sampling sequence (t_j) so that $(\Psi(t_j))$ is a discrete frame. Fornasier–Rauhut describe their contribution as a partial answer on source PDF pages 1–2 and give sufficient kernel-localization conditions; on source PDF page 31 they explicitly ask whether their L^1-integrability hypothesis is necessary [1].

Explicit later solution

Freeman–Speegle say in their abstract and introduction that they solve the discretization problem in full generality. Their Theorem 1.3, on supporting PDF pages 2–3, states the following [2].

Let (X, Σ) have measurable singletons and let $\Psi : X \to H$ be measurable. The following are equivalent:

1. There is a countable sequence (t_j) in X (repetitions allowed) such that $(\Psi(t_j))$ is a frame for H.
2. There is a positive σ-finite measure ν on (X, Σ) such that Ψ is a continuous frame with respect to ν and $\|\Psi(t)\|$ is bounded outside a ν-null set.

In particular every bounded continuous frame is sampleable. Theorem 1.4 strengthens this uniformly: there are universal $C, D > 0$ such that every continuous Parseval frame taking values in the Hilbert unit ball has a sampling with lower and upper frame bounds C, D.

The universal statement without boundedness is false. Freeman–Speegle give the continuous frame

$$\Psi(n) = \sqrt{n}\, e_n \quad (n \in \mathbb{N}), \qquad \mu(\{n\}) = 1/n,$$

for ℓ_2. Any sampling with dense span must contain every direction e_n, and its vector norms are then unbounded; a discrete frame is necessarily norm bounded. Thus this frame cannot be discretized.

Proof intuition

Normalize a bounded continuous frame by its frame operator to a Parseval frame in the unit ball. The Marcus–Spielman–Srivastava/Kadison–Singer partition machinery, assembled through finite-dimensional approximations, selects countably many frame vectors with universal two-sided frame bounds. Undoing the normalization yields a sampled frame for the original family. Conversely, if a sampling already exists, put counting mass (including multiplicity) on the sampled points. The same family is then an essentially bounded continuous frame for that atomic σ-finite measure. These two directions produce the exact criterion.

Provenance and scope

This is an explicit later solution, not an agent-generated theorem. It fully resolves when the underlying Hilbert continuous frame admits a sampled Hilbert frame. It does not assert that every sampling also yields the source's simultaneous atomic decompositions and Banach frames for all associated coorbit spaces; those stronger conclusions retain localization hypotheses. Nor does it settle the separate source remark asking whether the kernel algebra $\mathcal{A}_1$ is spectral.

References

[1] M. Fornasier and H. Rauhut, *Continuous Frames, Function Spaces, and the Discretization Problem*, arXiv:math/0410571v1 (2004).

[2] D. Freeman and D. Speegle, *The discretization problem for continuous frames*, arXiv:1611.06469 (2016), Theorems 1.3 and 1.4.